

Robotics Academy: New Power Tower Inspection Using Deep Learning

Program: GSoC 2026

Organization: JdeRobot

Project: #3

Project Size: Medium (~175h)

Duration: Jun – Sep 2026

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Demo Video: https://youtu.be/aZTyB_sHlI

Time Zone: UTC+3

Mentors: David Pascual-Hernández, Md. Shariar Kabir, Luis Roberto Morales

1. ABSTRACT

Power infrastructure inspection is a safety-critical task typically performed manually at significant cost and risk. This project proposes a new deep learning-based exercise in JdeRobot's RoboticsAcademy where students train and deploy a neural network that detects and classifies structural defects on power towers using a simulated drone as the inspection platform. Key outcomes include a Gazebo Harmonic simulation environment, a labeled defect dataset, a reference YOLOv8 model, an extended Simple API, and comprehensive Jupyter Notebook learning materials.

2. MOTIVATION

RoboticsAcademy already hosts a classical power-tower inspection exercise relying on colour thresholds and edge detection. While useful, it does not teach students to apply modern deep learning pipelines. Three problems motivate this project: - No deep learning exercise exists connecting drone perception with structured defect classification. - Students lack a reproducible simulation environment and labelled dataset to train their own models. - The existing Simple API has no standardised way to return structured DL outputs (bounding boxes, logits, confidence).

3. TECHNICAL PROPOSAL

3.1 Simulation Environment

A Gazebo Harmonic scene featuring power towers with physically modelled defects (corrosion, broken insulators, missing hardware). A drone with RGB camera will navigate pre-programmed waypoints — mirroring real UAV inspection workflows.

3.2 Dataset Collection & Labelling

Images recorded from Gazebo across multiple configurations and lighting conditions. Bounding-box annotations produced using CVAT, exported in YOLO and COCO formats. Dataset uploaded to HuggingFace Hub or Zenodo.

3.3 Deep Learning Model

Reference model trained using YOLOv8 (primary) and EfficientDet (secondary). Training tracked with MLflow. Target: $mAP@0.5 > 0.75$. Trained weights distributed alongside the exercise.

3.4 Exercise Integration

Follows architecture of existing End-to-End Visual Control exercise. Students implement a Python module with `execute()` function receiving a camera frame and returning structured predictions. RADI Docker image extended with new Gazebo world and HAL/GUI bindings.

3.5 Simple API Extension

Four new output fields to strictly follow JdeRobot's requirements:

- `set_boxes(boxes)` — bounding box coordinates
- `set_labels(labels)` — integer class indices
- `set_logits(logits)` — raw unnormalized predictions for classification
- `set_scores(scores)` — confidence values $[0,1]$

4. TIMELINE

Week 1-2 | Community Bonding — study codebase, mentor alignment

Week 3-4 | Gazebo power tower scene + drone navigation

Week 5-6 | Record dataset + label with CVAT

Week 7-8 | Train YOLOv8 model, evaluate with mAP/F1

Week 9 | MIDTERM — working defect detector + metrics

Week 10-11 | RoboticsAcademy exercise integration

Week 12-13 | Simple API extension (bbox, logits, confidence)

Week 14 | Jupyter Notebooks + dataset upload

Week 15-16 | Testing, final polish, submission

5. DELIVERABLES

- 1) Gazebo Harmonic power tower simulation environment
- 2) 500+ labeled image dataset (public HuggingFace/Zenodo)
- 3) Trained YOLOv8 model (mAP > 0.75)
- 4) Extended Simple API with bbox, logits + confidence support
- 5) Fully integrated RoboticsAcademy exercise
- 6) Jupyter Notebook tutorials and documentation

6. TOOLS & TECHNOLOGIES

Simulation: Gazebo Harmonic (or Classic, pending mentor alignment), ROS2 Humble, Aerostack2

Deep Learning: Python, PyTorch, YOLOv8, EfficientDet

Data: CVAT, Roboflow, HuggingFace

Tracking: MLflow

Infrastructure: Docker, HAL/GUI API, GitHub Actions

7. ABOUT ME

I am Mustafa Kerem Güçlü, a 2nd-year Computer Engineering student at Ankara University. During this GSoC application I successfully set up the RoboticsAcademy Docker environment and implemented a Follow Line solution featuring adaptive PID control with dynamic Kp adjustment, asymmetric brake/acceleration system, and real-time HUD visualization. Why this project? Power tower inspection sits at the intersection of drone robotics, computer vision, and deep learning — the exact areas I want to deepen. I am motivated by the social impact of safer infrastructure inspection and by creating learning materials for future engineering students. Commitment: I can dedicate 25-30 hours per week throughout GSoC. No conflicting internships or exams during the coding phase.

Demo Video: https://youtu.be/aZTyB_sHlal

GitHub: <https://github.com/keremmg/JdeRobot-GSoC-2026>

Github Name: keremmg

8. RISKS & MITIGATION

Gazebo complexity → Use simplified procedural towers, start data collection early

Gazebo version mismatch → Will align with mentors on whether to use Gazebo Harmonic or stick to the current Gazebo Classic baseline during community bonding.

Model below target → Fall back to pretrained COCO backbone, transfer learning

Docker integration issues → Follow existing exercise PRs, consult mentors early